Alisa Sheinkman

alisasheik@gmail.com ○ Website ○ LinkedIn ○ GitHub

Profile

Machine learning researcher and engineer with a PhD in Statistics, developing Bayesian learning algorithms, deep generative models and LLM orchestration systems. Research and production experience in Python, PyTorch and JAX, spanning scalable inference, uncertainty quantification and model evaluation.

Selected Project & Award

Bayesian LLM Orchestrator **Code**

Awesome Serverless award, Nebius Serverless AI Builders Challenge

- Built and evaluated an adaptive LLM routing and evaluation workflow using hierarchical Bayesian reliability models to select models and request second opinions under a token-cost objective.

Research & Engineering Experience

Research Associate, Imperial College London 2026 – Present

- Developed and trained a deep generative model of malaria transmission and drug resistance evolution.
- Investigated and improved simulation-based Bayesian inference methods.

Research Scientist, 40folds 2026

- Developed Bayesian media mix models and time-series forecasting algorithms, with robust validation frameworks for model evaluation.
- Implemented scalable, production-ready ML code using Python, JAX, PyTorch and GCP.

Doctoral Researcher in ML, University of Edinburgh 2020 – 2025

- Developed a novel Bayesian neural network architecture with shrinkage priors and derived scalable variational inference algorithms.
- Implemented methods in Python, JAX and NumPyro; compared architecture and inference choices through predictive accuracy, calibration and computational cost.

Selected Publications

A. Sheinkman, E. Semenova, S. M. Hall, T. J. Hitge, and J. Cockayne. **NeurIPS, 2025.**
SMRS: advocating a unified reporting standard for surrogate models in the artificial intelligence era.

A. Sheinkman and S. Wade. **ECML PKDD, 2025.**
Understanding the Trade-offs in Accuracy and Uncertainty Quantification: Architecture and Inference Choices in Bayesian Neural Networks.

Education

PhD in Statistics, University of Edinburgh 2020 – 2025

MASt in Mathematical Statistics (Merit), University of Cambridge 2019 – 2020

Bachelor of Mathematics (First Class Honours), NRU HSE 2015 – 2019

Technical Skills

Programming & ML	Python, PyTorch, JAX, NumPyro, NumPy, SciPy, pandas, scikit-learn
LLMs & agents	Transformers, Hugging Face, LangGraph, Bayesian model routing
Methods	Deep learning, generative modelling, variational inference, model evaluation
Infrastructure	GCP, Docker, Kubernetes, Git, GitHub

Miscellaneous

Publications and a full list of research activities are available on the website.